EVERY COHERENT SHEAF ON A NOETHERIAN ALGEBRAIC SPACE IS FILTERED WITH SUBQUOTIENTS PUSHED FORWARD FROM IDEAL SHEAVES ON IRREDUCIBLE SUPPORTS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Splitting a coherent sheaf along a union of two strictly smaller closed supports. Same chapter spaces-cohomology.tex, label lemma-prepare-filter-support (invoked at proof line 2565). It produces the short exact sequence $0 \to \mathcal{G}_1 \to \mathcal{F} \to \mathcal{G}_2 \to 0$ with supports in the two smaller closed pieces, which turns the reducible case into a contradiction with minimality.
- A power of the ideal of the support annihilates the sheaf. Same chapter spaces-cohomology.tex, label lemma-power-ideal-kills-sheaf (invoked at proof line 2584). It gives $\mathcal{J}^n\mathcal{F} = 0$, so that the filtration by the powers of $\mathcal{J}$ reduces the assertion to the case where $\mathcal{J}$ annihilates $\mathcal{F}$.
- The scheme theoretic support, so that the sheaf is a pushforward from the support. Morphisms of Spaces, label spaces-morphisms-lemma-i-star-equivalence (invoked at proof line 2600). It identifies the scheme theoretic support of $\mathcal{F}$ with T, so that $\mathcal{F}$ is the pushforward of a coherent module on T and the preparation lemma for the irreducible case applies.
- Preparation of a coherent module on a reduced irreducible space by a pushed forward ideal power. Same chapter spaces-cohomology.tex, label lemma-prepare-filter-irreducible (invoked at proof line 2601). It gives the short exact sequence $0 \to i_*(\mathcal{I}^{\oplus r}) \to \mathcal{F}

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\to \mathcal{Q} \to 0$ with $\mathcal{Q}$ supported in a proper closed subset of T, whose subquotients have the required shape.
- Noetherianity of the underlying topological space, and the reduced induced structure on a closed subset. Properties of Spaces, labels spaces-properties-lemma-Noetherian-topology (invoked at proof line 2554) and spaces-properties-lemma-reduced-closed-subspace (invoked at proof line 2582). The first supplies the minimal element of the collection of counterexample supports; the second provides the ideal sheaf defining the reduced structure on that support.

Source reference. Original lemma 07UK: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2339–2339. Navigation pointer for \ref{lemma-power-ideal-kills-sheaf} used inside the counted span (line 2584); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UQ: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2478–2478. Navigation pointer for \ref{lemma-prepare-filter-irreducible} used inside the counted span (line 2601); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UP: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2435–2435. Navigation pointer for \ref{lemma-prepare-filter-support} used inside the counted span (line 2565); the argument that consumes it is retained in the span.

Source reference. Original lemma 04CJ: sources/stacks-chow/tags.txt, source lines 5693–5693. Navigation pointer for \ref{spaces-morphisms-lemma-i-star-equivalence} used inside the counted span (line 2600); the argument that consumes it is retained in the span.

Source reference. Original lemma 04ZF: sources/stacks-chow/tags.txt, source lines 6463–6463. Navigation pointer for \ref{spaces-properties-lemma-Noetherian-topology} used inside the counted span (line 2555); the argument that consumes it is retained in the span.

Source reference. Original lemma 03IQ: sources/stacks-chow/tags.txt, source lines 4612–4612. Navigation pointer for \ref{spaces-properties-lemma-reduced-closed-subspace} used inside the counted span (line 2583); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. Let $\mathcal{F}$ be a coherent sheaf on X. There exists a filtration*

$$0 = \mathcal{F}_0 \subset \mathcal{F}_1 \subset \ldots \subset \mathcal{F}_m = \mathcal{F}$$

by coherent subsheaves such that for each $j = 1, \ldots, m$ there exists a reduced closed subspace $Z_j \subset X$ with $|Z_j|$ irreducible and a sheaf of ideals $\mathcal{I}_j \subset \mathcal{O}_{Z_j}$ such that

$$\mathcal{F}_j/\mathcal{F}_{j-1} \cong (Z_j \to X)_*\mathcal{I}_j$$

Proof. Consider the collection

$$\mathcal{T} = \left\{ \begin{array}{c} T \subset |X| \text{ closed such that there exists a coherent sheaf } \mathcal{F} \\ \text{with } \mathrm{Supp}(\mathcal{F}) = T \text{ for which the lemma is wrong} \end{array} \right\}$$

We are trying to show that $\mathcal{T}$ is empty. If not, then because $|X|$ is Noetherian (Properties of Spaces, Lemma 04ZF) we can choose a minimal element $T \in \mathcal{T}$. This means that there exists a coherent sheaf $\mathcal{F}$ on X whose support is T and for which the lemma does not hold. Clearly $T \neq \emptyset$ since the only sheaf whose support is empty is the zero sheaf for which the lemma does hold (with $m = 0$).

If T is not irreducible, then we can write $T = Z_1 \cup Z_2$ with Z_1, Z_2 closed and strictly smaller than T. Then we can apply Lemma 07UP to get a short exact sequence of coherent sheaves

$$0 \to \mathcal{G}_1 \to \mathcal{F} \to \mathcal{G}_2 \to 0$$

with $\text{Supp}(\mathcal{G}_i) \subset Z_i$. By minimality of T each of $\mathcal{G}_i$ has a filtration as in the statement of the lemma. By considering the induced filtration on $\mathcal{F}$ we arrive at a contradiction. Hence we conclude that T is irreducible.

Suppose T is irreducible. Let $\mathcal{J}$ be the sheaf of ideals defining the reduced induced closed subspace structure on T, see Properties of Spaces, Lemma 03IQ. By Lemma 07UK we see there exists an $n \geq 0$ such that $\mathcal{J}^n\mathcal{F} = 0$. Hence we obtain a filtration

$$0 = \mathcal{I}^n\mathcal{F} \subset \mathcal{I}^{n-1}\mathcal{F} \subset \ldots \subset \mathcal{I}\mathcal{F} \subset \mathcal{F}$$

each of whose successive subquotients is annihilated by $\mathcal{J}$. Hence if each of these subquotients has a filtration as in the statement of the lemma then also $\mathcal{F}$ does. In other words we may assume that $\mathcal{J}$ does annihilate $\mathcal{F}$.

Assume T is irreducible and $\mathcal{J}\mathcal{F} = 0$ where $\mathcal{J}$ is as above. Then the scheme theoretic support of $\mathcal{F}$ is T, see Morphisms of Spaces, Lemma 04CJ. Hence we can apply Lemma 07UQ. This gives a short exact sequence

$$0 \to i_*(\mathcal{I}^{\oplus r}) \to \mathcal{F} \to \mathcal{Q} \to 0$$

where the support of $\mathcal{Q}$ is a proper closed subset of T. Hence we see that $\mathcal{Q}$ has a filtration of the desired type by minimality of T. But then clearly $\mathcal{F}$ does too, which is our final contradiction. $\square$